

SARYA, Russian Federation

WMO: 272710

Lat: 58.367N

Lon: 45.533E

Elev: 125

StdP: 99.83

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.0	-28.4	-34.6	0.1	-31.8	-30.9	0.2	-28.3	7.8	-7.1	7.0	-7.2	1.0	340	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	29.0	20.3	27.0	19.2	25.2	18.1	21.3	27.2	20.3	25.4	19.3	23.7	2.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.4	23.8	18.5	13.5	22.7	17.5	12.7	21.8	62.5	27.4	58.8	25.5	55.5	23.7	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	6.2	5.4	-35.7	31.3	3.0	2.1	-37.9	32.8	-39.6	34.0	-41.3	35.2	-43.5	36.7		
(5)			-35.8	22.7	3.0	1.5	-37.9	23.7	-39.7	24.6	-41.4	25.4	-43.6	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-11.9	-10.5	-4.5	3.9	10.6	14.9	17.6	14.8	9.7	3.6	-3.6	-8.8		
	DBStd	11.69	8.29	7.95	5.50	5.28	5.09	4.23	3.70	3.96	4.28	4.63	6.02	8.05		
	HDD10.0	3216	679	573	450	194	57	8	1	5	56	203	407	582		
	HDD18.3	5648	938	807	708	434	245	119	59	123	261	457	657	841		
	CDD10.0	681	0	0	0	10	74	154	236	154	47	5	0	0		
	CDD18.3	72	0	0	0	0	4	16	37	14	2	0	0	0		
	CDH23.3	848	0	0	0	3	71	190	418	157	11	0	0	0		
	CDH26.7	211	0	0	0	0	13	46	106	45	1	0	0	0		
(14) Wind	WSAvg	2.6	2.6	2.7	2.8	2.8	2.8	2.4	2.0	2.1	2.3	2.8	2.7	2.8		
(15) Precipitation	PrecAvg	645	43	33	33	34	50	87	78	70	54	61	55	47		
	PrecMax	873	84	71	77	89	113	165	187	149	119	131	140	80		
	PrecMin	431	22	13	5	9	18	17	10	23	11	16	6	17		
	PrecStd	95	14	15	17	20	21	39	47	29	24	28	27	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.6	2.7	10.2	22.8	28.3	30.8	31.9	31.0	24.7	17.6	9.3	5.1		
		MCWB	1.0	1.4	4.8	12.5	17.3	21.0	21.2	21.5	16.8	11.9	8.0	4.7		
	2%	DB	0.8	1.5	6.0	19.0	25.2	27.7	29.3	27.1	21.5	13.9	6.4	2.7		
		MCWB	0.3	0.7	2.3	10.9	16.1	19.2	20.6	19.9	15.4	10.1	5.6	2.0		
	5%	DB	0.1	0.6	4.0	15.6	22.5	25.4	27.5	24.6	18.9	11.7	4.6	1.2		
		MCWB	-0.3	0.0	1.2	8.9	14.2	17.9	19.9	18.2	14.2	9.4	3.9	0.6		
	10%	DB	-1.1	-0.2	2.6	12.6	20.1	23.4	25.7	22.5	16.8	9.8	3.1	0.4		
		MCWB	-1.6	-0.8	0.7	7.4	12.6	17.1	19.0	17.4	13.2	8.4	2.5	0.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.8	5.1	13.2	19.3	22.0	22.7	22.5	17.9	12.9	8.1	4.9		
		MCDB	1.4	2.3	8.7	20.7	26.3	27.6	28.3	29.1	21.9	15.6	9.0	5.2		
	2%	WB	0.4	0.8	3.2	11.4	17.2	20.5	21.5	20.7	16.3	11.0	5.8	2.2		
		MCDB	0.7	1.3	5.2	17.7	22.6	26.3	27.1	26.0	20.0	13.2	6.2	2.6		
	5%	WB	-0.3	0.2	1.9	9.7	15.4	19.2	20.6	19.3	14.9	9.7	4.0	0.7		
		MCDB	0.1	0.7	3.3	14.7	20.4	23.7	26.0	23.3	18.3	11.2	4.5	1.0		
	10%	WB	-1.7	-0.8	1.0	7.9	13.8	17.9	19.8	18.0	13.5	8.4	2.5	0.0		
		MCDB	-1.2	-0.3	2.3	11.8	18.3	22.0	24.5	21.6	16.3	9.9	3.0	0.4		
(35) Mean Daily Temperature Range	MDBR		6.9	8.2	9.5	10.3	12.0	11.4	11.9	10.7	9.2	6.0	4.2	5.5		
	5% DB	MCDBR	4.1	4.8	10.1	15.1	15.9	14.9	14.8	14.4	12.8	9.6	5.0	3.6		
		MCWBR	4.1	4.4	7.0	8.1	7.6	7.2	6.8	7.1	7.1	6.2	4.6	3.6		
	5% WB	MCDBR	4.2	4.5	7.0	13.9	13.5	12.5	12.9	12.4	10.5	8.2	4.6	3.4		
		MCWBR	4.2	4.4	5.1	7.9	7.1	6.6	6.4	6.5	6.1	5.7	4.5	3.4		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.310	0.345	0.394	0.368	0.366	0.397	0.400	0.357	0.337	0.300	0.263		
	taud		2.057	2.057	2.041	2.129	2.397	2.424	2.350	2.320	2.422	2.353	2.178	2.025		
	Ebn at Noon		525	687	775	805	865	871	830	796	779	674	501	407		
	Edh at Noon		54	87	119	130	106	105	111	107	83	66	48	40		
(44) All-Sky Solar Radiation	RadAvg		0.41	1.13	2.44	3.79	5.21	5.45	5.45	4.14	2.42	1.05	0.38	0.23		
	RadStd		0.07	0.30	0.33	0.45	0.40	0.45	0.51	0.39	0.31	0.17	0.05	0.05		

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.56	N/A	N/A	N/A	N/A	N/A	-207	N/A

Nomenclature: See separate page